

INFO 703
BIG DATA AND ANALYTICS

HOTEL RESERVATIONS
ASSESSMENT ONE

HARIN RAMJI
21521277

Table Of Contents

Title -----	Page 1
Table of Contents-----	Page 2
Introduction-----	Page 3
Dataset Attributes-----	Page 4
Preprocessing -----	Page 5-6
Statistical Summary of Dataset -----	Page 8-9
Graphs & Analysis -----	Page 10-18
Classification & Analysis -----	Page 19-20
Conclusion -----	Page 21
References -----	Page 22

Introduction

In the era of big data, the ability to analyse and classify large datasets is very important for getting valuable/precious insights especially for a hotel. This report focuses on data analysis and classification using the "Classify Hotel Reservations" dataset sourced from Kaggle.

The Kaggle dataset is created by a user called Unmoved(<https://www.kaggle.com/unmoved>) who seems to have gotten the raw data from a GitHub repository and then cleaned up the data and assembled it to a proper format and csv file(https://github.com/alef-s/INN_hotels). The GitHub raw data is a mess with uneven datatypes and unmoved has cleaned up the data. I have further cleaned up and pre-processed the data to make it more suitable for analysis using WEKA and MS Excel.

The dataset contains important and valuable information on hotel reservations, including booking details, customer preferences, and cancellation patterns and we shall add our own columns later on for further analysis and drawing conclusions based on those calculated columns.

The primary objective of this study is to apply different classification techniques to this dataset, assess their performance, and determine the most effective model. The dataset is linked below and we shall also be using weka which is also linked below.

<https://www.kaggle.com/code/unmoved/classify-hotel-reservations/input>

<https://ml.cms.waikato.ac.nz/weka>

https://github.com/alef-s/INN_hotels

Why This Dataset Was Chosen

The "Classify Hotel Reservations" dataset was selected for several reasons. One of them being that I have met this problem before but from a business management perspective for my uncle's' vast Airbnb portfolio and Motels across the coast in Kenya. He had me working on the marketing side of the problem but as soon as I found a hotel reservation dataset, I took it since I have experience with this business-case before. My previous experience was streamlining the business process but now I can look at it at a statistical and mathematical problem.

Moreover, the dataset presents a real-world classification problem that can be tackled using machine learning techniques. Identifying patterns in customer reservations can help hotels improve their booking policies and increase profitability. Lastly, the dataset contains a mix of numerical and categorical attributes, making it a suitable candidate for looking at different data preprocessing techniques and classification models.

Possible Uses

By analysing this dataset, we can conclude on some business decisions. With the help of WEKA and MS Excel, we can run the business more efficiently and run advertisement campaigns accordingly. At the end of the report, I shall give some suggestions to the business to make more profits and streamline the business even further.

Dataset Attributes

The dataset consists of multiple attributes about what data they collect from reservations/bookings which we shall use for analysis later on. Below is a detailed explanation of each attribute and its datatype and format i.e. **(Nominal, Integer)**. The highlighted fields at the bottom are made in excel for analysis and shall be further discussed in the Preprocessing section of the report

Attribute	Description	Type
Booking_ID	A unique identifier for each booking.	Nominal (String)
No_of_Adults	The number of adults in the reservation.	Numeric (Integer)
No_of_Children	The number of children in the reservation.	Numeric (Integer)
No_of_weekend_nights	Number of weekend nights (Saturday & Sunday) included in the stay.	Numeric (Integer)
Type_of_meal_plan	The meal plan chosen by the customer. 1 - Breakfast ONLY 2 - Half Board (Breakfast + One meal) 3 - Full Board (All meals) 0 - No Meal Plan	Numeric (Integer)
Required_Car_Parking_Space	Whether the customer requested a parking space (1 = Yes, 0 = No)	Binary (Integer)
Room_Type_Reserved	The type of room reserved by the customer.	Numeric (Integer)
Lead_Time	Days between the booking date and arrival date.	Numeric (Integer)
Arrival_Year	The year of arrival for the reservation.	Numeric (Integer)
Arrival_Month	The month of arrival for the reservation.	Numeric (Integer)
Arrival_Date	The specific day of the month arrival.	Numeric (Integer)
Market_Segment_Type	The market segment to which the booking belongs (e.g., Online, Corporate, Offline, etc.).	Numeric (Integer)
Repeated_Guest	Whether the guest has previously stayed at the hotel (1 = Yes, 0 = No).	Binary (Integer)
No_of_previous_cancellations	The number of past cancellations made by the customer.	Numeric (Integer)
No_of_previous_bookings_not_canceled	The number of previous bookings by the customer that were not cancelled.	Numeric (Integer)
Avg_price_per_room	The average price per room for the reservation (fixed formatting issue in Excel).	Numeric (Float)
No_of_special_requests	The number of special requests made by the customer.	Numeric (Integer)
Booking_Status	Whether the reservation was cancelled (0 = Not Cancelled, 1 = Cancelled).	Binary (Integer)
Arrival Date Full	Convert the individual dates to a full date	Nominal (String)
Day of the week	Day of the week of reservation	Nominal (String)

Preprocessing

Preprocessing refers to the set of steps performed on raw data to clean, transform, and structure it before analysis or modelling. It ensures that data is accurate, consistent, and in a suitable format for machine learning or statistical analysis.

Class Attribute Selection

For preprocessing in this dataset, I have looked through the data and found NO empty fields or NULL/Error fields which are easy to spot in MS excel.

The average price per room is stored as a string instead of a number and needs some changes.

'\$65.00' this is how it was saved including the quotes and a simple find and replace removed all the quotes and dollar signs in the whole attribute. The new field now looks like below

rev no	of_prev	avg_price	no_of_spec
0	0	65.00	0
0	0	106.68	1
0	0	60.00	0
0	0	100.00	0
0	0	94.50	0
0	0	115.00	1

For complementary customers the avg_price_per_room was stored as a hyphen "-" which was prompting weka to assume the field is a string. A quick find and replace helped me remove all the hyphens and changed the values to 00.00

F	F
type_of_meal_plan	type_of_meal_plan
'Meal Plan 1'	1
'Not Selected'	0
'Meal Plan 1'	1
'Meal Plan 1'	1
'Not Selected'	0
'Meal Plan 2'	2

The **type_of_meal_plan** column contained categorical values such as 'Meal Plan 1', 'Meal Plan 2', etc. These were converted into numeric values:

- 'Meal Plan 1' → 1
- 'Meal Plan 2' → 2
- 'Meal Plan 3' → 3
- 'Not Selected' → 0

H	H
room_type_reserved	room_type_reserved
'Room_Type 1'	1
'Room_Type 1'	1
'Room_Type 1'	1
'Room_Type 1'	1
'Room_Type 1'	1
'Room_Type 1'	1
'Room_Type 1'	1
'Room_Type 4'	4

The **room_type_reserved** column, which originally contained nominal string values, was converted into numerical form:

- 'Room Type 1' → 1
- 'Room Type 2' → 2
- 'Room Type 3' → 3
- 'Room Type 4' → 4
- 'Room Type 5' → 5
- 'Room Type 6' → 6
- 'Room Type 7' → 7

S	S
booking_status	booking_status
Not_Canceled	0
Not_Canceled	0
Canceled	0
Canceled	1
Canceled	1
Canceled	1
Not_Canceled	1

The **booking_status** column, originally labelled as 'Not_Canceled' and 'Cancelled', was converted into a binary format:

- 'Not_Canceled' → 0
- 'Cancelled' → 1

M	M
market_segment_type	market_segment_type
Offline	0
Online	1
Online	1
Online	1
Online	1

The Market_segment_type column originally had 'Offline', 'Online', 'Corporate', 'Complementary' Or 'Aviation' and was converted to integer format

- 'Offline' → 0
- 'Online' → 1
- 'Corporate' → 2
- 'Complementary' → 3
- 'Aviation' → 4

Creating a new column for Arrival date full using the formula below

=DATE(J2,K2,L2)

This produced this kind of information. the format is DD/MM/YYYY

T
Arrival Date full
02/10/2017
06/11/2018
28/02/2018
20/05/2018
11/04/2018
13/09/2018
15/10/2017

I also created another column for the day of the week it is for the arrival date with the formula below

=TEXT(T2, "dddd")

And it created this type of a field which will help us in analysis.

U
Day of the week
Monday
Tuesday
Wednesday
Sunday
Wednesday
Thursday

These transformations helped convert categorical variables into a structured numerical format, making the dataset more suitable for analysis.

STATISTICAL SUMARRY OF THE DATASET

Attribute	Minimum	Maximum	Mean (Average)	Standard Deviation
BookingID	1	36275	18138	10471.835
No_of_Adults	0	4	1.845	0.519
No_of_children	0	10	0.105	0.403
No_of_weekend_nights	0	7	0.811	0.871
No_of_week_nights	0	17	2.204	1.411
Type_of_meal_plan	0	3	0.95	0.48
Required_car_parking_space	0	1	0.031	0.173
Room_type_reserved	1	7	1.709	1.4
Lead_time	0	443	85.233	85.931
Arrival_year	2017	2018	2017.82	0.384
Arrival_month	1	12	7.424	3.07
Arrival_date	1	31	15.597	8.74
Market_Segment_Type	0	4	0.797	0.616
Repeated_guest	0	1	0.026	0.158
No_of_previous_cancellations	0	13	0.023	0.368
No-of_previous_bookings_not_cancelled	0	58	0.153	1.754
Avg_price_for_room	0	540	103.424	35.089
No_of_special_requests	0	5	0.62	0.786
Booking status	Distinct: 2			
Arrival Date Full	Distinct: 549			
Day of the week	Distinct: 7			

The table provides statistical findings into the dataset using key metrics such as **minimum, maximum, mean (average), and standard deviation** for each attribute. Below is an interpretation of these values and what they reveal about the dataset:

Understanding Standard Deviation (SD)

Standard Deviation measures how much the data varies from the mean. A **high standard deviation** indicates that the values are spread out over a wider range. A **low standard deviation** suggests that the values are closely clustered around the mean.

Interpretation of Key Attributes

Lead Time (SD = 85.931)

The mean lead time is **85.233 days**, but with a high standard deviation of **85.931**. This means that while some bookings are made far in advance, others are made at the last minute. Example: Some guests may book **6 months ahead (180+ days)**, while others book just **a day before arrival (0 days)**.

The **minimum, maximum, and mean (average)** provide important information about the range of each attribute. Let's look at the importance of each and how they help in understanding the dataset.

- **Minimum (Min):** The smallest observed value in the dataset.
- **Maximum (Max):** The largest observed value in the dataset.
- **Mean (Average):** The sum of all values divided by the number of records

These values help in:

- Understanding the range (spread) of data.
- Identifying extreme values (outliers).
- Analysing typical values (mean).

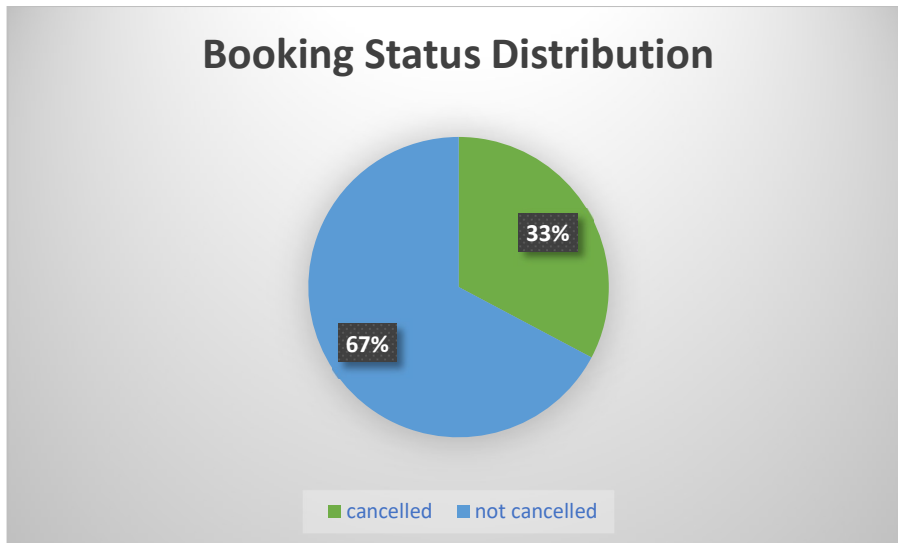
For Example, here are some interpretations below:

(a) No_of_Adults --Min: 0 (There exist some bookings with no adults, possibly errors or children's reservations). **Max:** 4 (Maximum number of adults in a booking). **Mean:** 1.845 (Most bookings have **1 or 2 adults**). Most bookings are made for small groups or couples, while larger groups (4 adults) are less common.

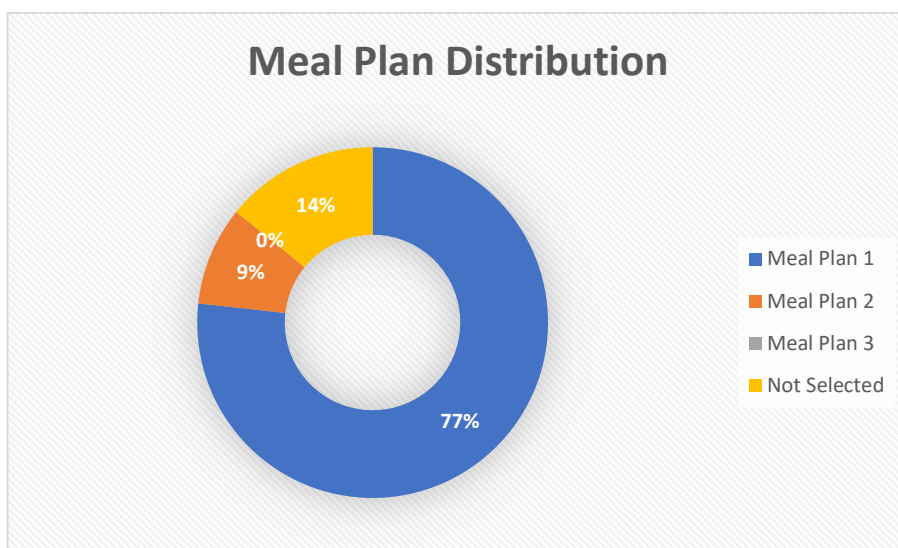
(b)Lead-time -- Min: 0 (Some bookings are made on the same day as check-in). **Max:** 443 (Some bookings are made more than **a year in advance**). **Mean:** 85.233 (**Most bookings are made approximately 3 months in advance**). Some guests book at the last minute (**same day**), Others plan way ahead (**over a year**).

GRAPHS AND ANALYSIS

Let's now look at some graphs and try to understand them and hopefully suggest some ideas to the hotel management on what to do to increase profits. Some of these graphs are made by the help of Microsoft Excel while some are made by WEKA. MS Excel requires creating of pivot tables so that I can make some complex graphs.



The pie chart above shows that **33%** of the bookings are cancellations. To reduce this, the business should analyse the causes, streamline processes, and improve customer experience to encourage more successful transactions. One of the causes, I believe, is a free cancellation policy that means that people abuse a lot of and implementing a cancellation charge shall reduce these numbers significantly.



The pie chart above represents the proportion of hotel reservations that selected different meal plans.

Meal Plan 1 Dominates (77%) The vast majority of guests opted for Meal Plan 1, indicating that it is the most preferred option. This could be due to affordability, variety, or inclusiveness in certain booking packages.

Meal Plan 3 (9%) and Meal Plan 2 (0.01%) Are Less Popular. Meal Plan 3 has some traction but is significantly less popular than Meal Plan 1. Meal Plan 2 has an extremely low selection rate (0.01%), suggesting it is almost irrelevant to customers and most likely needs to be removed.

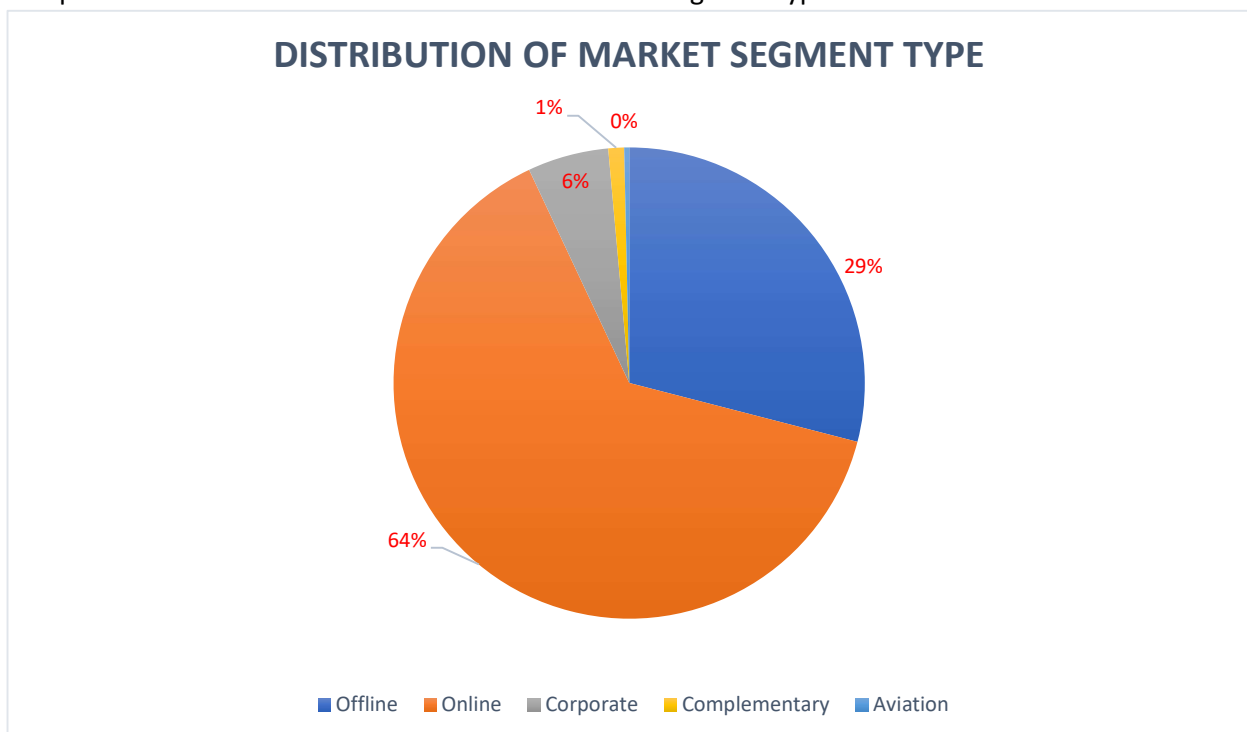
A Considerable Number of Guests Didn't Select Any Meal Plan (14%). Some guests may prefer dining outside or bringing their own food. This group presents an opportunity to upsell meal plans during check-in or the booking process.

Suggestion for the Hotel:

Reconsider Meal Plan 2: Since almost no one selects it, the hotel should either redesign the plan (adjust pricing or offerings) or remove it altogether to simplify choices.

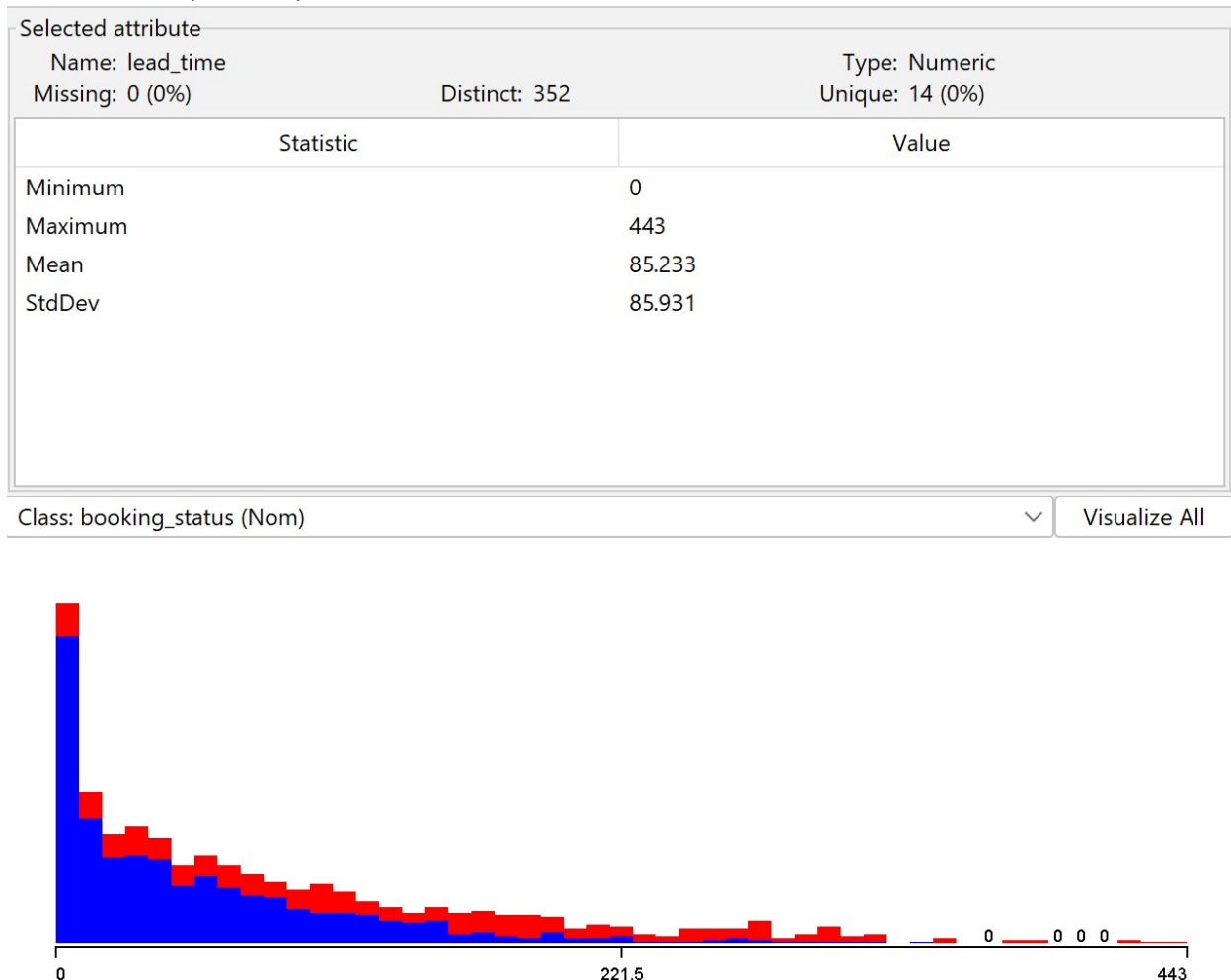
Encourage Meal Plan Selection at Booking: Since 14% of guests didn't select a meal plan, the hotel could introduce discounts for pre-booking or bundle meal plans with room rates to increase adoption.

The pie chart below shows the distribution of Market Segment type

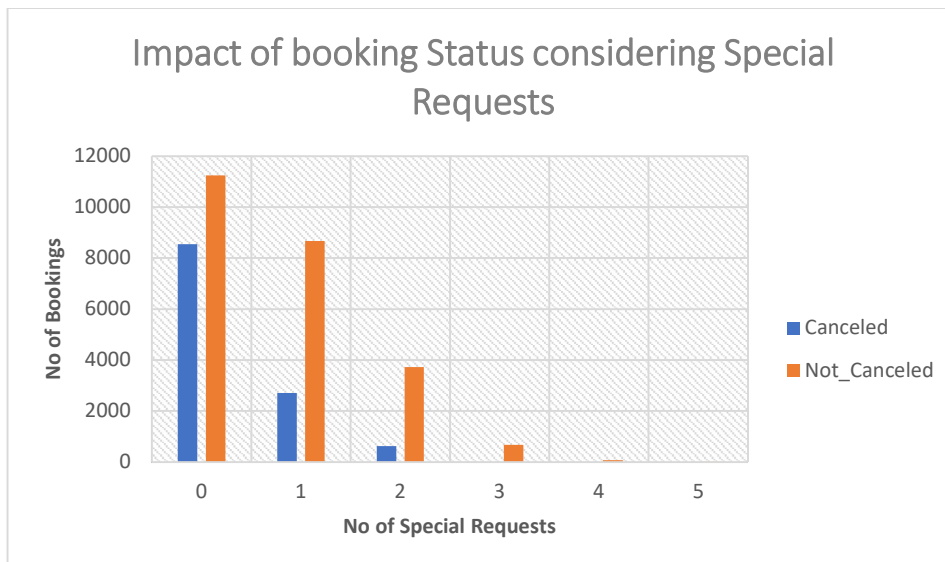


It clearly shows that online bookings are the most popular followed by offline bookings. The hotel management could boost revenue by investing in easier bookings online or some kind of discount or promotion for online bookings. The offline bookings are most probably made by the older generation who do not like/are unable to book online.

Below is a Histogram of lead time generated by WEKA. Lead time is the time in days between the booking date and the arrival date. As you can see the maximum is 443 days meaning a party booked a room almost a year early.



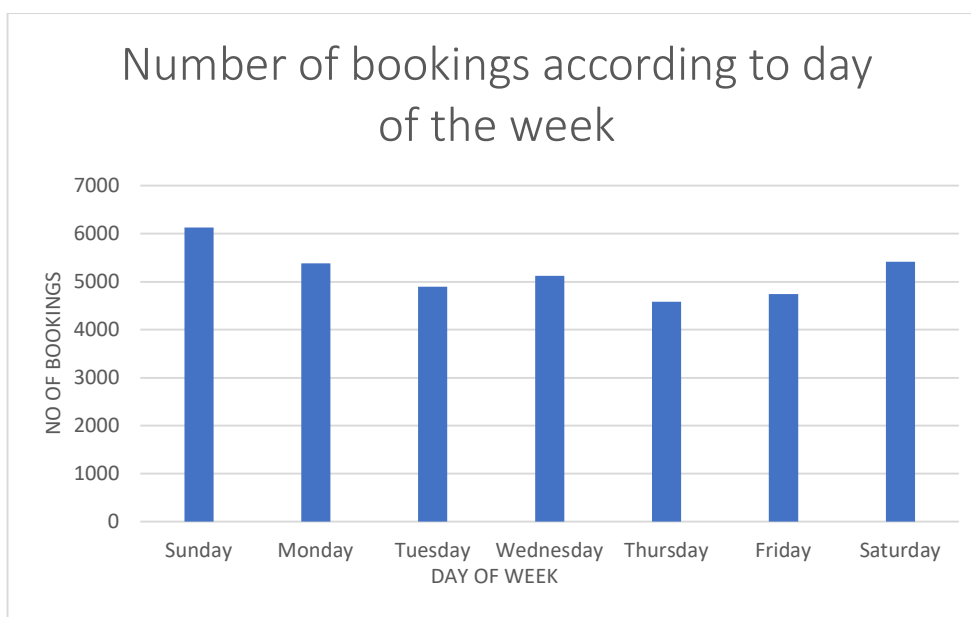
Since the histogram indicates that most bookings happen last minute or within a month of the arrival date, hotels could implement dynamic pricing strategies. For example, offering discounts for early bookings or imposing stricter cancellation policies for last-minute reservations might help reduce cancellations or also imposing high prices for last minute bookings.



This clustered Bar chart above clearly shows that as number of special requests increase the likelihood of cancellation reduces significantly. This means that the number of special requests is a good indicator of commitment and can therefore be given special promotions to keep them happy and loyal. Guests with no special requests may be more price-sensitive and more likely to cancel when cheaper alternatives arise. Personalization (meeting guest requests) might increase perceived value, reducing cancellations.

This means that if a special request is made, we can be confident the booking is not cancelled and so we can focus on retaining them as a regular reliable customer by offering special service or discounts for future bookings.

The graph below shows the number of bookings made according to the day of the week and we shall analyse it later. I have also made a heat map so that we can clearly quantify which is the busiest days and which days are the slowest.



Row Labels	Count of Booking_ID
Sunday	6135
Monday	5380
Tuesday	4899
Wednesday	5120
Thursday	4581
Friday	4741
Saturday	5419
Grand Total	36275

Peak Booking Days (Sunday, Saturday, and Monday)

Sunday is the day of the week with the highest number of bookings, with 6135 reservations, suggesting strong demand on that particular day of the week. This could be attributed to customers checking in for weekend getaways or extended stays.

Saturday (5419 bookings) follows closely, meaning that the weekends attract the highest number of guests. Many customers may prefer Saturday check-ins for short vacations.

Monday (5380 bookings) also shows a high volume of reservations, potentially due to business customers arriving for the workweek.

Midweek Low (Tuesday to Thursday)

Tuesday (4899 bookings), Wednesday (5120 bookings), and Thursday (4581 bookings) experience a noticeable decline in reservations compared to weekends.

Thursday has the lowest number of bookings, suggesting it is the least preferred check-in day. This drop could be due to fewer customers and/or business guests typically arriving earlier in the week or simply put it as preference and cannot influence this decision.

Weekend vs. Weekday Trends

Weekends (Saturday & Sunday) see significantly more bookings than weekdays (Monday–Friday), highlighting a strong preference for travel over the weekend or family short vacations over the weekend. Since the hotel is near the beach this could be short vacations near the beach where customers arrive on Saturday and leave on Sunday. Weekdays still maintain steady bookings, with Monday being the busiest, likely due to corporate travel.

Suggestions to Management

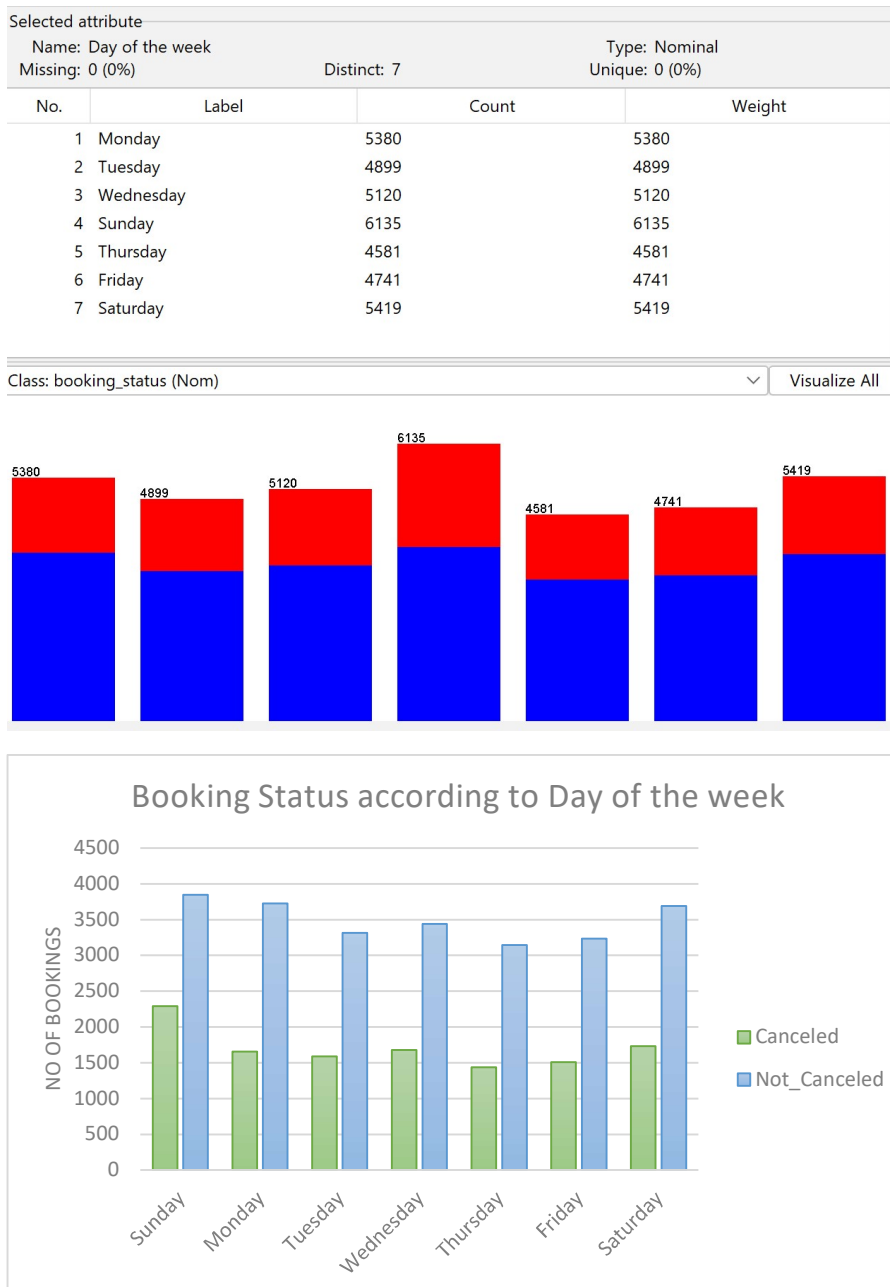
Revenue Optimization: Increase prices on high-demand days (Sunday, Saturday, Monday) and offer discounts on low-demand days (Tuesday–Thursday) to maximize revenue.

Marketing & Promotions: Run midweek deals (e.g., free breakfast, spa offers) and target business customers with conference access and flexible check-ins. Promote weekend getaway packages for leisure customers/Families.

Operational Efficiency: Adjust staff schedules to match demand, reinforcing housekeeping and front desk teams on peak booking days (Sunday).

Below is, also, a histogram of the day of the week the arrival. However, we now want to visualise how many of those bookings were cancelled vs Not cancelled. The previous 2 graphs (Heatmap and bar chart) showed total bookings but now we want to see the cancellation trends according to the day of the week.

I have also made the graph in MS Excel to visualise it better. The histogram shows that Sunday is the most popular day but is also the most cancelled day as well but the bar chart below it is much more visual and is easier to draw conclusions on.



The graph represents the **Booking Status according to the Day of the Week**, showing the number of bookings that were either **Cancelled** (green bars) or **Not Cancelled** (blue bars) for each day.

Analysis:

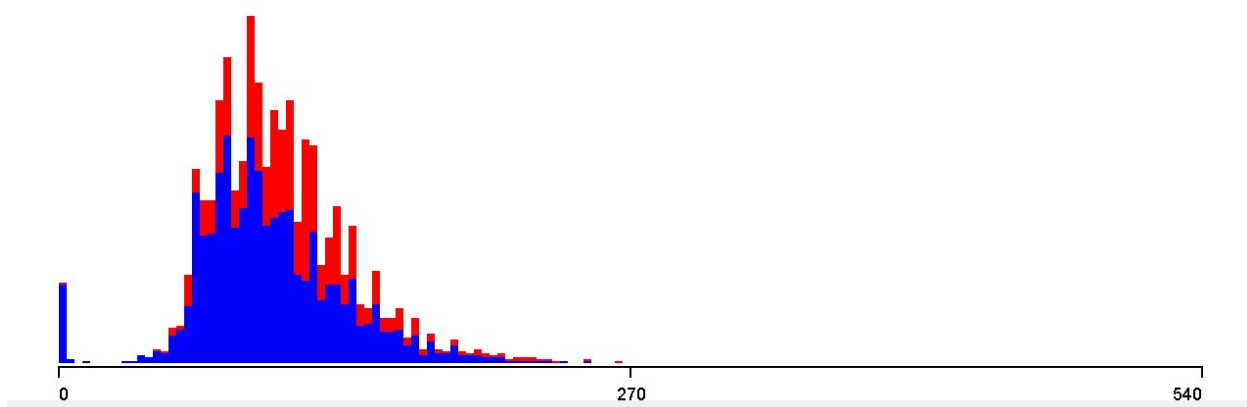
Trend of Not Cancelled Bookings (Blue Bars) -- The number of Not Cancelled bookings remains consistently high throughout the week. The highest values appear on Sunday, Monday, and Saturday. The lowest value is on Friday, but it's still significantly higher than cancelled bookings.

Trend of Cancelled Bookings (Green Bars) -- The cancelled bookings are relatively lower compared to the ones that were not cancelled. The highest cancellation rate appears on Sunday and Wednesday but they also have the highest bookings compared to the rest of the week so it looks like a byproduct. The more the bookings the higher the chance of cancellations. However, Thursday and Friday show the lowest number of cancelled bookings.

- Higher bookings on weekends: This could indicate increased demand for reservations during weekends, possibly for leisure or tourism purposes.
- Stable non-cancelled bookings: Many people may book and commit to their reservations regardless of the day.

Selected attribute		
Name: avg_price_per_room	Type: Numeric	
Missing: 0 (0%)	Distinct: 3930	Unique: 1991 (5%)
Statistic		Value
Minimum		0
Maximum		540
Mean		103.424
StdDev		35.089

Class: booking_status (Nom) Visualize All



This histogram above, generated using WEKA, visualizes the distribution of the average price per room along with cancellation frequencies. The majority of rooms are priced below the average, while a few high-end, premium rooms exist. The column at 0 represents complimentary stays offered free of charge. Additionally, the

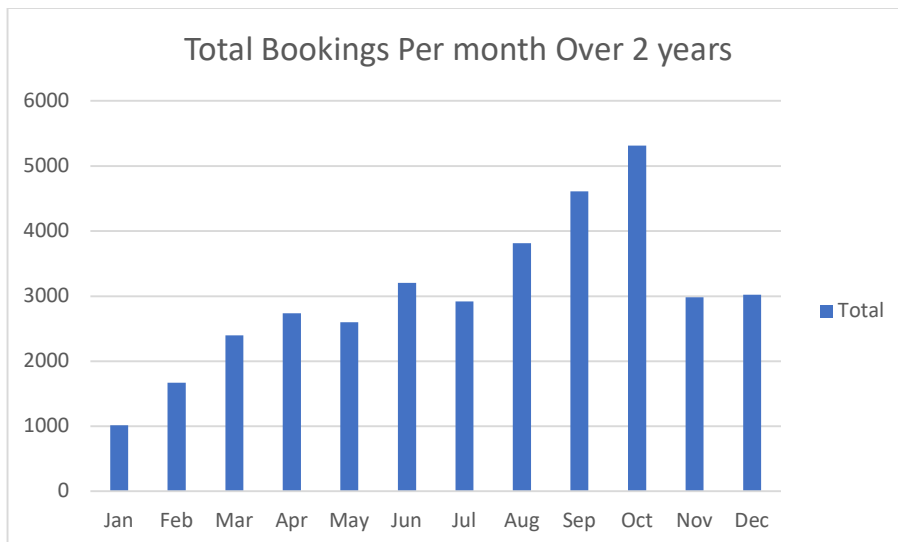
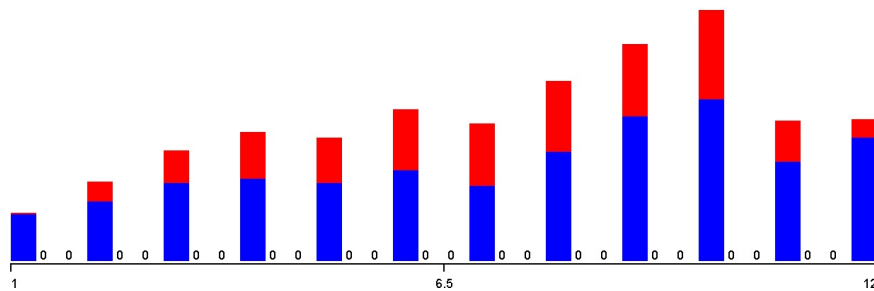
mid-range priced rooms appear to be the most frequently booked and, consequently, also have the highest cancellation rates.

The hotel should consider a slight increase in room prices across all categories to boost overall revenue. Additionally, implementing incentives such as discounts for non-refundable bookings or early payment options for mid-range rooms could help reduce cancellations while maintaining high occupancy.

The histogram below shows the monthly trend of bookings and its cancellations (highlighted in red) and a bar chart showing total bookings for every month of the year which we shall later draw conclusions on.

Selected attribute		
Name: arrival_month		Type: Numeric
Missing: 0 (0%)	Distinct: 12	Unique: 0 (0%)
Statistic	Value	
Minimum	1	
Maximum	12	
Mean	7.424	
StdDev	3.07	

Class: booking_status (Nom) Visualize All



The data reveals clear **seasonal trends**, where certain months experience higher bookings, while others see a lower turnout of reservations.

Peak Season (August, September, October)

The highest number of bookings occur in August, September, and October, meaning those are the hotel's peak months. This aligns with Kenya's tourism high season, driven by favourable weather, international tourist arrivals, and events like the Great Migration in the Maasai Mara, which attracts many travellers/tourists who may also seek coastal relaxation.

Beachfront locations in Kenya are particularly attractive during these months as visitors look to combine safari adventures with a beach getaway and also experience some unique adventures near the coast. Higher cancellation rates during these months could indicate increased competition among tourists securing multiple accommodations.

Low Season and Off-Peak Months

The data shows a decline in bookings during the rainy season (April to June), which is common for coastal hotels in Kenya due to unpredictable weather and reduced tourist inflows.

Off-peak months present an opportunity to attract local travellers, corporate retreats, and discounted package tourists to maintain occupancy.

Business Recommendations after analysing monthly trends

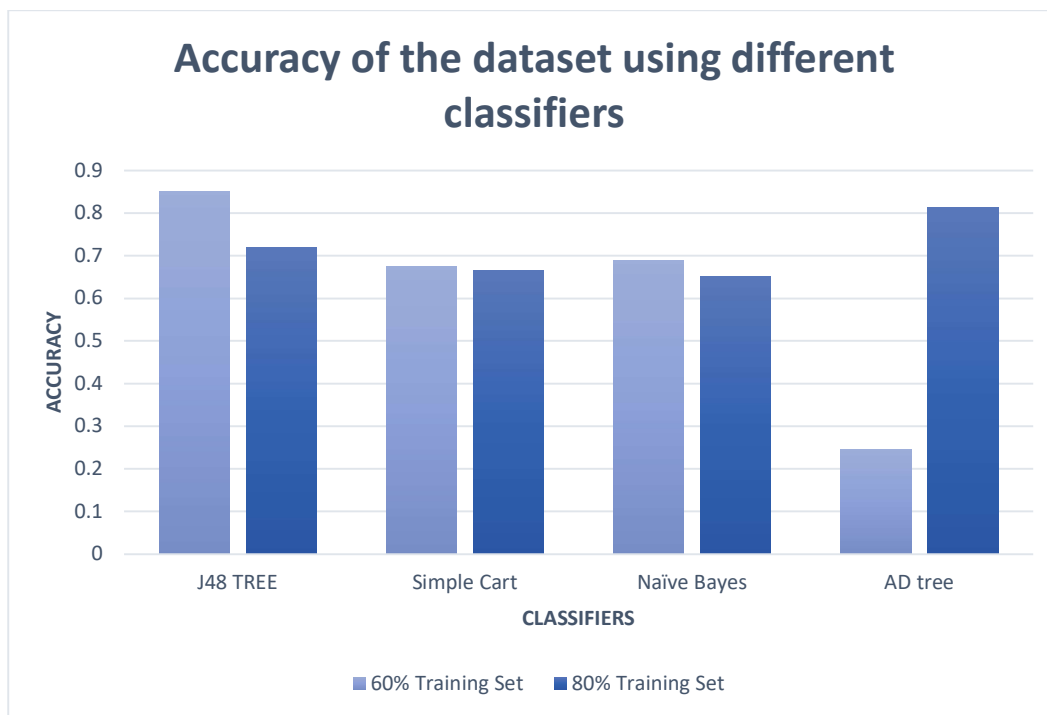
Seasonal Pricing Strategy -- Increase room prices during peak months (August to October) to maximize revenue while ensuring high-value services for premium tourists. Offer discounted packages in the low season (April to June) to attract domestic travellers, corporate events, and long-stay guests.

Targeted Advertisement & Promotions -- Promote the hotel as a relaxation destination/ water sports destination during peak tourism months. Run off-season campaigns for local visitors, offering discounts, free upgrades, or bundled experiences (e.g., Jet skis, Hot air balloons, water sports, or Hiking/trekking the mountainous regions of Kenya). Reduce cancellations by offering discounts for early bookings and flexible rebooking options.

Operational Adjustments -- Increase staffing and resources during peak months to handle high guest volumes efficiently. Scale down staffing and inventory during low-demand months to reduce operational costs without compromising service quality.

CLASSIFICATION

	Training set size	TP rate	FP rate	Accuracy
J48 TREE	60%	0.851	0.190	0.851
	80%	0.859	0.185	0.720
Simple Cart RepTree	60%	0.675	0.675	0.675
	80%	0.666	0.666	0.666
Naïve Bayes	60%	0.664	0.213	0.689
	80%	0.652	0.218	0.652
AD tree	60%	0.791	0.325	0.245
	80%	0.813	0.250	0.813



Based on the performance metrics above and the graph (Bar chart) of the accuracy against the classifiers, the **J48 decision tree** is the best classifier for my dataset. The reasons for choosing the J48 Tree classifier are listed below

1. Accuracy

The J48 tree achieves the highest accuracy across both training set sizes. **60% Training Data: 85% Accuracy & 80% Training Data: 85% Accuracy.** Other classifiers, such as RepTree and Naïve Bayes, have significantly lower accuracy, with values around 66-68%.

2. True Positive (TP) Rate

The True Positive rate measures the proportion of instances that were classified correctly. J48 has the highest TP rate (**0.851 for 60% training, 0.859 for 80% training**), meaning it correctly identifies more actual bookings/cancellations than other models.

3. False Positive (FP) Rate

A lower False Positive rate indicates fewer incorrect classifications of non-cancellations as cancellations. J48 has an FP rate of (**0.190 for 60% training and 0.185 for 80% training**), which is lower than AD Tree (0.325) and comparable to Naïve Bayes. It, however, is significantly lower than Simple Cart meaning it is way much better than Simple Cart.

4. Stability Across Training Sizes

Unlike some classifiers that drop in performance when increasing the training size (e.g., AD Tree with an accuracy of only 24.5% at 60% training but 81% at 80% training size, J48 remains stable and reliable, improving slightly when trained on more data.

Since **J48 offers the highest accuracy, best TP rate, and a low FP rate**, it is the most suitable classifier for predicting hotel booking cancellations. It balances precision and recall effectively, making it the best choice for your dataset.

Summary of Findings and Recommendations

This report provides an analysis of hotel reservation data of the beach front hotels and Airbnb's in Mombasa, identifying trends, customer behaviour, and opportunities for revenue optimization. Our insights are based on data visualizations created using Microsoft Excel and WEKA, highlighting booking patterns, cancellations, meal plan preferences, lead time trends, special requests, market segments, and pricing strategies.

Key Findings & Strategic Recommendations

❖ High Cancellation Rate (32.76%)

A huge proportion of bookings (32.76%) are cancelled, which impacts revenue/profits. The management should implement stricter cancellation policies, such as charging a small cancellation fee to discourage whimsical bookings.

❖ Meal Plan Preferences & Revenue Optimization

Meal Plan 1 dominates (76.73%) the meal plan preferences, while Meal Plan 2 is nearly irrelevant (0.01%) and should be reconsidered or removed. 14% of guests do not select any meal plan, representing an opportunity for the hotel to act on it.

Remove/reduce price for Meal Plan 2 and introduce meal plan incentives, such as discounted pre-booked meal packages to increase guests choosing any of the meal plans. The ultimate goal is to reduce/ discourage guests not choosing a meal plan which would mean less revenue for the hotel.

When booking online preselect Meal plan 1 as an option such that customers would have to opt-out of a meal plan, there could be a possibility that the customer has forgotten to choose a meal plan during booking.

❖ Market Segment & Booking Channels

Online bookings are the most popular, followed by offline bookings. Enhance the interface of the online booking systems and introduce exclusive discounts or promotions for online reservations to encourage direct bookings and reduce dependency on third-party platforms. Make it easier to book online but try not to discourage the offline booking then the hotel loses out on the older generation customers who rely on offline bookings.

❖ Lead Time & Pricing Strategy

While some guests book far in advance (up to 443 days in advance), most bookings occur last-minute or within a month of arrival. Implement dynamic pricing, where early bookings receive discounts while last-minute bookings are priced higher. Customers with a higher lead time might get a lower/no cancellation fee but customers with a lower/last minute booking should get a cancellation fee since it has caused the most inconvenience to the business.

❖ Special Requests as an Indicator of Commitment

Guests making special requests are significantly less likely to cancel, which shows a higher commitment level. Reward guests with special requests by offering loyalty perks, or future booking discounts to enhance guest satisfaction and the hotel may just earn a repeat customer.

❖ Weekly Booking Trends & Revenue Strategies

Increase room rates on high-demand days (Sunday, Saturday, Monday) to increase revenue which will in turn increase profits. Offer weekday discounts (Tuesday–Thursday) to attract more guests. Run weekend getaway promotions targeting tourists. Strengthen cancellation policies for Sunday bookings due to its high cancellation rate.

❖ Room Pricing & Cancellation Trends

Mid-range rooms are booked the most but also have the highest cancellations. Higher-end rooms have lower cancellation rates.

Slightly increase room prices across all categories while introducing discounts for non-refundable or early-payment bookings to reduce cancellations.

❖ Seasonal Trends & Tourism Patterns

Peak season: August, September, October, driven by international tourism and the Great Migration in Kenya. Low season: April–June, correlating with the rainy season.

Increase room rates during peak months to increase revenue. Introduce special packages/discounts for the off-season to try and recuperate some of the costs.

Conclusion

By analysing this dataset and drawing conclusions given above Implementing dynamic pricing, stricter cancellation policies, targeted promotions, and operational adjustments will significantly improve the revenue and profits of the Hotels. Properly implementing updated cancellation policies will help reduce cancellations, but if executed/implemented poorly, it could drive customers away by making them feel unfairly treated.

REFERENCES

Dataset

<https://www.kaggle.com/code/unmoved/classify-hotel-reservations/input>

https://github.com/alef-s/INN_hotels

Application used

<https://ml.cms.waikato.ac.nz/weka>

Author of dataset

<https://www.kaggle.com/unmoved>